UMHACKATHON 2026 · Z.AI TRACK

ClaimIQ

AI-first medical claim adjudication.

From 7 days to 7 seconds.

FastAPI + Python | Gemini 2.0 Flash | ilmu.ai GLM | RAG + FAISS | Supabase + SQLite

Third Party Administrator Platform · Medical Fraud Detection · Bilingual EOB

THE PROBLEM

Your mother is in hospital. Her claim is number 1,847.

She has Dengue Hemorrhagic Fever. The clinic submits her RM 1,650 claim. A team of 12 human reviewers reading claims one by one — hers takes 3 to 7 working days. The hospital deposit counter does not wait.

7

Days

Average manual processing time

5·5×

Cost

More expensive than automated

$40B

Fraud

Lost globally per year

THE SOLUTION

What is ClaimIQ?

An AI-powered Third Party Administrator platform that adjudicates medical claims in real-time — the way a senior clinical reviewer would, but in seconds, not days.

Reads medical documents
Doctor notes, lab reports, X-rays, PDFs

Cross-checks insurance policy
RAG engine retrieves only the relevant policy rules

Detects fraud
Compares doctor notes against actual lab document values

Makes adjudication decisions
Approved · Denied · Referred — with full audit trail

Human safety net built in
Borderline cases always go to a human reviewer with full AI analysis

ARCHI TECTURE

The Double-Agent System

Two specialist AIs — not one. Because reading a document and reasoning about it are fundamentally different cognitive tasks.

AGENT 1 — VISION

Gemini 2.0 Flash

Reads handwritten notes, scanned lab reports, X-rays, PDFs. Converts them into clean structured JSON data.

PERCEPTION ONLY — does not reason

JSON

AGENT 2 — ADJUDICATION MAIN BRAIN

ilmu.ai GLM

Malaysian AI model. Receives structured data + policy rules from RAG engine. Performs clinical reasoning and fraud scoring.

DECISION — Approved / Denied / Referred

RAG ENGINE (FAISS Vector DB)

Retrieves only the relevant policy rules for each diagnosis — not the whole policy document. Prevents hallucination.

THE PIPELINE

10-Step Clinical Pipeline

01 Intake

GP submits claim — patient, IC, clinic, diagnosis, evidence files

02 Scrubbing

Duplicate detection and data integrity checks

03 Eligibility

Real-time member coverage verification via registry

04 Processing

Vision Agent (Gemini) extracts clinical findings from documents

05 Policy Retrieval

RAG (FAISS) fetches only the relevant policy rules for this diagnosis

06 Adjudication

ilmu.ai GLM performs medical reasoning against retrieved policy

07 Fraud Scoring

Risk-matrix flags: name mismatch, value discrepancy, billing anomalies

08 Safety Freeze

Borderline claims gated — human reviewer makes final call

09 Audit Logging

Immutable step-by-step trace stored in PostgreSQL

10 Closing

Bilingual Explanation of Benefits generated (EN + BM)

FRAUD DETECTION — LIVE CASE

You cannot lie to a system that reads your own documents.

CLAIM SUBMITTED

Patient: Siti Nurhaliza · IC: 880505-10-5555

Diagnosis: Acute Dengue Hemorrhagic Fever (A91)

Amount: RM 1,650 · Tanglin Medical Center

Doctor states: Platelet 15 x10^3/uL · Hematocrit >55%

NS1 Antigen: POSITIVE · Admission required

VS

LAB REPORT (READ BY VISION AGENT)

Patient on report: MUHAMMAD KHUDRI BIN HALIM BASHAH

← DIFFERENT PERSON

Platelet count: 176 x10^3/uL ← COMPLETELY NORMAL

Hematocrit: 37.1% ← COMPLETELY NORMAL

No critical values detected

3 FRAUD ALERTS FIRED AUTOMATICALLY

⚠ FRAUD ALERT

Identity mismatch

Lab report belongs to a different patient entirely

⚠ FRAUD ALERT

Platelet fabricated by >1,000%

Doctor stated 15, actual lab shows 176 x10^3/uL

⚠ FRAUD ALERT

Hematocrit fabricated by ~50%

Doctor stated >55%, actual lab shows 37.1%

Result: DENIED · Fraud Risk: HIGH · Confidence: 0.92

DESIGN PHILOSOPHY

The Human Safety Net

ClaimIQ does not replace human judgment — it handles what does not need it, and supercharges what does.

AUTO-APPROVED

Clean claims

Eligibility verified. Evidence matches. Amount within limit. No flags. No missing documents.

The AI handles it fully. No human needed. Seconds, not days.

AUTO-DENIED

Clear violations

Unregistered patient. Duplicate submission. Missing mandatory document.

Denied with a full written explanation and exact policy reference.

REFERRED

Everything else

Any ambiguity. Any borderline case. Any situation the AI is not fully confident about.

Human reviewer gets the full AI analysis and makes the final call.

A patient's life can depend on getting this decision right. The AI accelerates the obvious. Humans own the edge cases.

Claims Intelligence Dashboard

Dashboard | Claims Queue | Denials | Fraud | Analytics | GP Portal | Submit

61.5%
Clean Claim Rate
Target >85%

15.4%
Denial Rate
Target <5%

20.4
Avg AR Days
Target <40 days

100%
Auto-Adjudicated
Target >80%

545
Total Claims

335
Approved

84
Denied

52
High Risk Flags

RM 140
Avg Claim

RM 42.4k
Total Approved

CLAIM STATUS COLOURS

- Approved — AI handled, no human needed
- Denied — clear policy violation
- Referred — human reviews with full AI analysis
- Intake — new claim just entered
- Error — document unreadable
- Appealing — doctor fighting the denial

AI WEEKLY INTELLIGENCE REPORT

This week the TPA processed 545 claims with an approval rate of 61.5% and a denial rate of 15.4%. The average claim was RM 140.08. Fraud screening flagged 52 high-risk claims. Auto-adjudication remained at 100%. Top clinic: Klinik Kesihatan Taman Melati (82 claims).

Generate with GLM

Real AI-generated text — not hardcoded

Six Pages. Every one functional.

Claims Queue

· Every claim with ID, date, clinic, diagnosis, amount, status

· Real Malaysian patient names + ICD-10 codes

· Colour-coded status badge — instant visual scanning

· Duplicate claims visible side-by-side

Denial Management

· CARC codes with frequency counts

· CARC 4 & 45 both at 14 — duplicate billing & benefit breach

· Every denied claim shows 'Appeal with GLM' button

· GLM generates a formal appeal letter in seconds

Analytics & Intelligence

· Every clinic ranked by denial rate, fraud flags, avg amount

· Klinik 1Malaysia Kepong: 26.9% denial rate flagged

· TPA manager can intervene before it becomes a crisis

· Intelligence that doesn't exist in most Malaysian TPAs

GP Clinic Portal

· Doctor logs in, sees only their own claims and statuses

· English / Bahasa Malaysia toggle — bilingual interface

· REFERRED status shown clearly — AI not sure, human reviewing

· One-click appeal letter generation via GLM

Submit a Claim

· Patient name, IC, clinic, date, diagnosis, ICD-10, amount

· Free-text clinical narrative — doctor writes their notes

· Upload supporting evidence files (lab reports, X-rays)

· System cross-references text vs documents — catches fraud

Fraud Dashboard

· 52 high-risk flagged claims visible in demo

· Each flag shows exactly what was detected and why

· Patient name mismatch, value discrepancy, billing anomaly

· Full traceable reason — not a black-box decision

TECHNI CAL STACK

Built to not crash. Built to scale.

BACKEND

FastAPI (Python)

Asynchronous — handles many claims simultaneously without blocking. Every claim processed in parallel.

DATABASE

Supabase (PostgreSQL) + SQLite

Cloud primary. Auto-falls back to local SQLite if internet drops. Demo never crashes — by design.

VISION AI

Google Gemini 2.0 Flash

State-of-the-art multimodal model. Reads images, PDFs, text in a single pass. GEMINI_API_KEY env var.

REASONING AI

ilmu.ai GLM

Malaysian language model — understands local healthcare context and BM medical terminology. GLM_API_KEY.

KNOWLEDGE BASE

FAISS Vector Database

Medical policy indexed as vector embeddings. Only retrieves relevant policy rules per claim — not the whole document.

FRONTEND

Vanilla JS + CSS

No frameworks. Every line understood. Loads fast. No dependency vulnerabilities. Full dark mode UI.

KEY FILES: api_server.py · database.py · claims_processor.py · adjudication_agent.py · vision_agent.py · rag_engine.py

VISION & REAL-WORLD IMPACT

Faster decisions. Fairer system.

As confidence scores grow, a patient admitted at 2am gets their pre-authorisation answered before they wake up.

PATIENT

Decision before the doctor finishes the admission note

Bilingual EOB in English and Bahasa Malaysia so patients actually understand what was paid and why

CLINIC / GP

Faster payment, less paperwork

Appeal letters generated in seconds. Denial reasons explained clearly. No more guessing why a claim was rejected.

TPA / INSURER

100% auto-adjudication with full audit trail

Fraud caught at intake. Immutable PostgreSQL logs for every decision. Analytics showing which clinics need attention.

THE SYSTEM

Lower fraud = lower premiums for everyone

Faster access = more Malaysians who can use their insurance when they actually need it. That is the real win.

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ClaimIQ

Claim number 1,847 should not have to

We built a system that gives back the hours doctors waste on forms, the days patients spend waiting, and the trust the insurance system is supposed to earn but rarely does.

- **10-Step Clinical Pipeline**
- **Double-Agent AI Architecture**
- **Cross-Reference Fraud Engine**
- **Human Safety Net Built In**
- **Bilingual EOB (EN + BM)**
- **100% Auto-Adjudicated Demo**

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